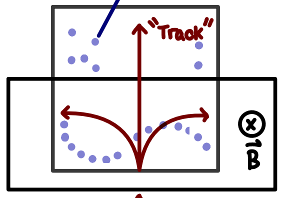
particle physics

3 branches
- solidstate — condensed matter
- particle
- astronomy/cosmology

① Cloud chamber
+ radioactive material
+ magnetic



$\vec{F} = q\vec{v} \times \vec{B}$
↳ + or −

why spirals?
- particle collides with aerosolized H_2O, loses energy via lower $\vec{v}$

② What keeps the protons in the nucleus? →

typical energy

p ⟷ p

radioactive Source

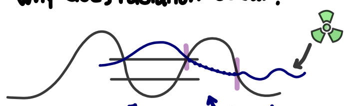
$U \sim \frac{e^2}{(4\pi\varepsilon_0)r} \sim \frac{1.44 \times 10^{-7}\ \text{eV-cm}}{10^{-13}\ \text{cm}} \sim 10^6\ \text{eV} = \text{MeV}$ where

$\frac{e^2}{4\pi\varepsilon_0} = 1.44 \times 10^{-7}\ \text{eV-cm}$

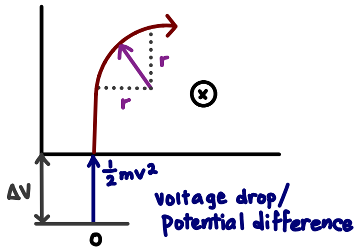
$r \sim 10^{-15}\ \text{m} = 10^{-13}\ \text{cm}$

$k_B T \sim 25\ \text{meV}$

why does radiation occur?



"mass spectrometer"

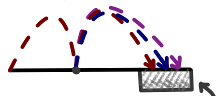


equate centrifugal force to magnetic force

$F_\perp = \frac{mv^2}{r} = qvB$

$q\Delta V = \frac{1}{2}mv^2 \rightarrow mv^2 = 2q\Delta V$

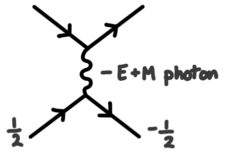
$\hookrightarrow r^2 = \frac{m^2v^2}{q^2B^2} = \frac{m(2q\Delta V)}{q^2B^2}$

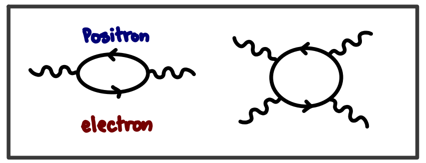


Can identify different substances via trajectory

$r^2 = \left(\frac{m}{q}\right)\frac{2\Delta V}{B^2}$ (m)

'strong force' – (recall charge / E&M Force)
– bosonic field





'fermionic' feynmann diagrams

fermions

	q	I	II	III
quarks	2/3 e	up	charm	top
	−1/3 e	down	strange	bottom
leptons	−1	electrons	muons	tau
	0	e-neutrino	μ-neutrino	τ-neutrino

bosons
- gluons – strong nuclear
- photons – E+M
- $W^{\pm}$-boson, Z^0-boson – weak nuclear

'hadrons' – anything made of quarks
- $B\bar{B}$ mesons — even # quarks, particle anti-particle pair
- baryons RGB — odd # quarks, even charge (fermions)

nucleons are composite particles like atoms
↳ made of 'quarks' – fermions, spin $\frac{1}{2}$

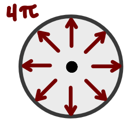
° color charges: RGB → together neutralize to 'white': All 3 RGB → white
anti-color/color pair: $B\bar{B}$ → white

"fine-tuning"
- entropy
- unitless ratios of physical constants

e.g. ratio of gravitational force to electric force: $\frac{F_G}{F_E} = \frac{\frac{Gm_e^2}{F}}{\frac{k_e e^2}{F}} = \sim 10^{-45}$

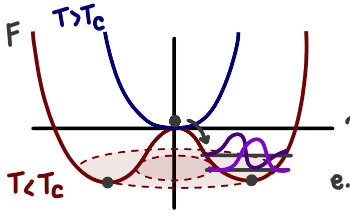
aestheticism of physics as a consequence of geometry

↳ why not: $F_G = \frac{Gm_e^{1.999}}{F}$?



↳ inverse square law
intensity – $\frac{S}{\text{area}} = \frac{S}{4\pi r^2}$

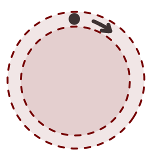
Spontaneous symmetry breaking – Higg's Boson
↳ quantization of oscillation in SSB potential well



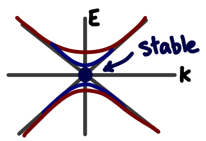
$\psi = |\psi_0| e^{i\phi}$ "order parameter"

e.g. $F \propto |\psi|^4$ T>Tc
$\propto |\psi|^4 - |\psi|^2$ T<Tc

Goldstone Boson



$2\pi r = 2N$



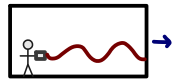
stable oscillation mode of field itself = Higg's Boson

weak anthropic principle — We always will find ourselves in a universe we can exist within

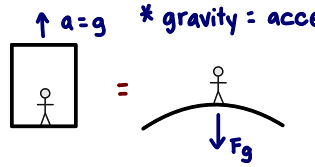
Strong anthropic principle — 'Boltzmann Brain'

final anthropic principle
- theism – god made it so
- deism – we are God and we made it so

General Relativity

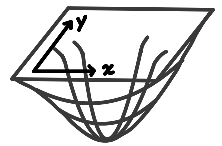


Special Relativity
Equal Reference Frame



* gravity = acceleration *

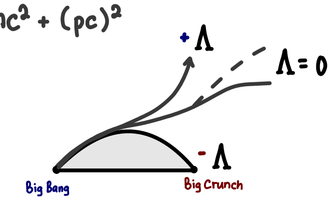
4 dimensional



'Geodesics' – all forces can be related to a 'curvature field' on a spherical shell where forces relate to deformations on the curvature of the matrix field, expressed as:
'curvature field': $mc^2 + (pc)^2$

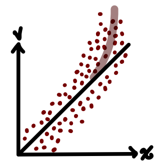
ρ, $\vec{E}$, ψ, g } what appears as a straight line or shortest path to an observer can in reality be a geodesic path along curved space-time

Free Parameter – Cosmological Constant Λ
If Λ is set to zero → universe expands



Doppler Shift – Velocity of objects relative to us

Hubble



→ LeMaitre – closed universe
recent data – open universe

+Λ – Big Rip/Freeze [Heat Death]
Repulsive (accelerated expansion)

Λ = 0 – Depends on matter density
Neutral (decelerated expansion)

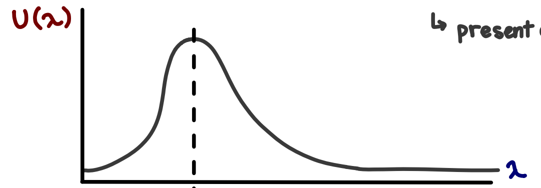
−Λ – Big Crunch
Attractive (contraction)

Cosmology – highest
- hot → expand volume
- colder → static/shrink

e.g. decoupling of E+M Field from quark/matter fields

Particle – smallest

→ Photon Planck distribution
→ continues to expand + cool
↳ present day T ~ 3k



"Cosmic microwave background radiation"